Algorithm for solving Sliding Puzzle

$A^*(\underset{S_0}{\text{start}}, \underset{G}{\text{goal}})$

1. Use the invariant to decide whether G is achievable from S_0.
2. Set $g[S_0] \leftarrow 0$; $\text{parent}[S_0] \leftarrow \text{Nil}$; $\text{Move}[S_0] \leftarrow \text{Nil}$
3. Enqueue S_0 into priority queue Q.

4. while $Q \neq \phi$

5. Dequeue - Min Q to obtain state S

6. If S = goal G then return the states and moves using Parent and Move (stops)

7. Generate every legal successor V with corresponding blank move m.

8. For every succesd V, set

newCost $\leftarrow g(S)+1$

9. If $v \notin$ g-table or newCost $< g(v)$

then $g[v] \leftarrow$ newCost,

parent$[v] \leftarrow s$,

move$[v] \leftarrow m$

10. Compute $h(v)$ and enqueue or update v in Q with

$f(v) = g(v) + h(v)$.

Manhattan Distance $A(r_1, c_1)$, $B(r_2, c_2)$

$$d = d(A, B) = |r_1 - r_2| + |c_1 - c_2|$$

Sliding Puzzle → No of rows and columns seperating current position from the goal

eg:

S_0

1	2	3
⑨	4	6
7	5	8

A(2,1) - position of 9 in S_0

B(3,3) - position of 9 in Goal

$$d_9 = d(A,B) = |2-3| + |1-3|$$

$$= 1 + 2 = 3$$

(Not taken for $d(S_0)$)

$$d(S_0) = d_1 + d_2 + d_3 + d_4 + d_5 + d_6 + d_7 + d_8$$

$$= 0 + 0 + 0 + 1 + 1 + 0 + 0 + 1$$

(↑ cal. using Manhattan)

$$= \underline{3}$$

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Priority Queue Q is ordered by tuple

$(f(s), h(s), s)$

- $f(s)$: smallest preferred
- $h(s)$: smaller h is preferred
- s: lexicographically smallest (makes tied runs reproducible - not needed for optimality)

States: $S_i[f, h]$

S_0

1	2	3
⑨	4	6
7	5	8

123946758

G

1	2	3
4	5	6
7	8	⑨

$I(S_0) = +1$

$I(G) = +1$

} Goal is achievable

$g[S_0] = 0$

parent $[S_0]$ = Nil

Move $[S_0]$ = Nil

Q:

S_0: $f(S_0) = g + h = 0 + 3 = 3$

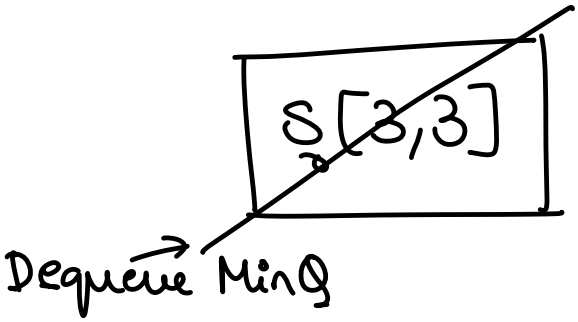
$h = d(S_0) = 3$

$S_0[f, h] = S_0[3, 3]$

Q $\boxed{S_0[3,3]}$

(Enqueue S_0 into Q)

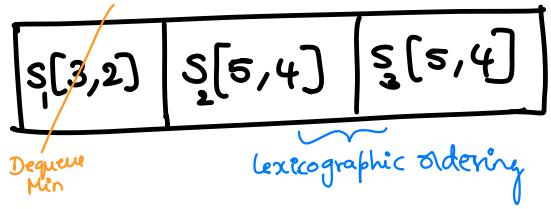
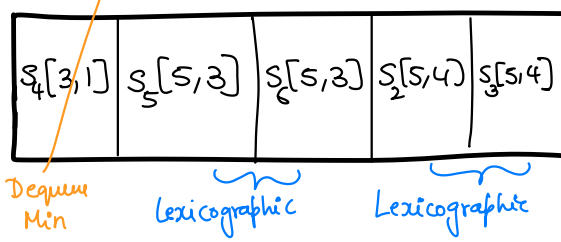
$Q \neq \phi$

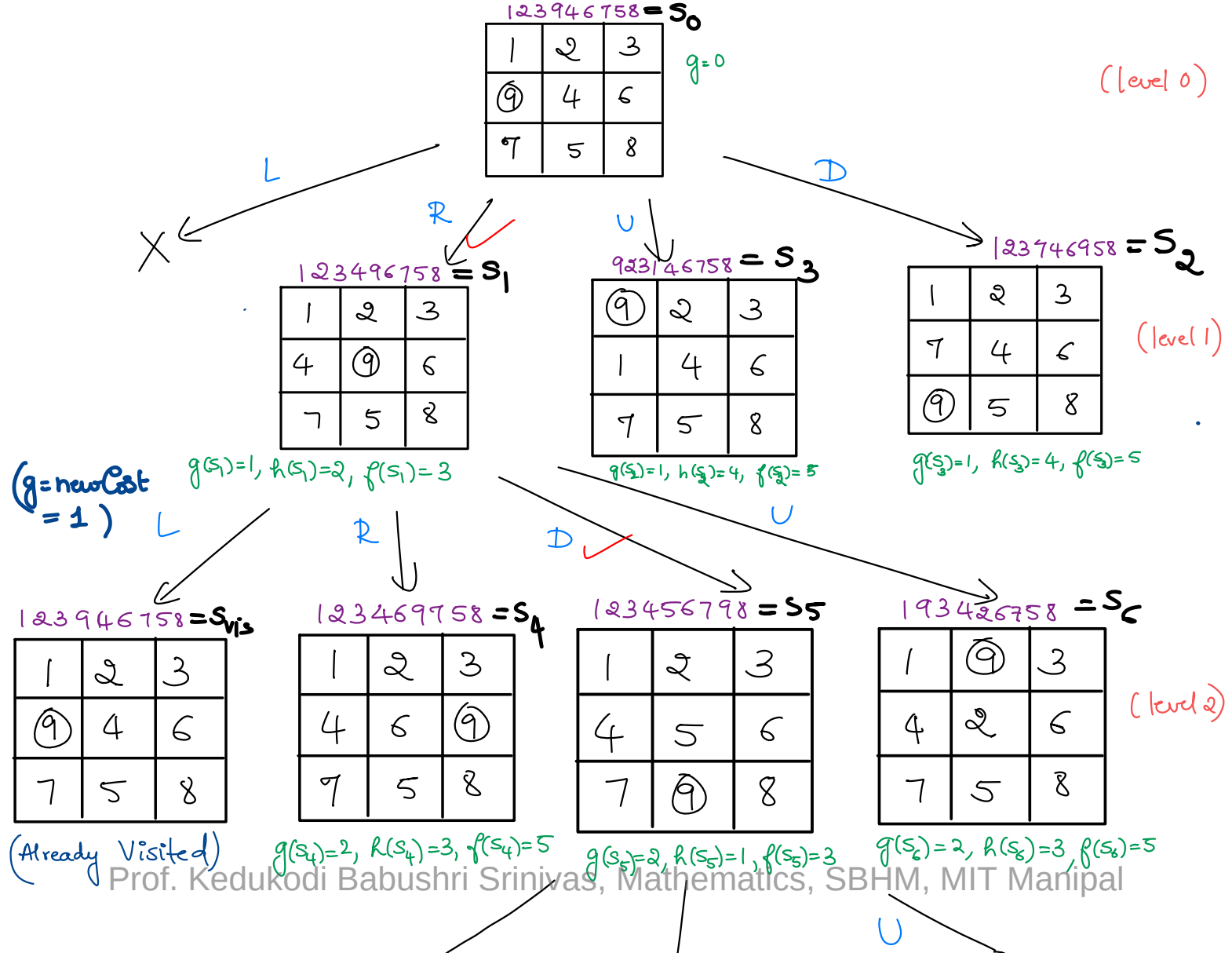


(Dequeue MinQ)

$S = S_0[3,3]$

$S \neq$ goal G

Level of Step	Dequeue-Min	Newly Enqueued States	Q after updating in Lexicographic order
0	$S_0[3,3]$	$S_1[3,2], S_2[5,4]$ $S_3[5,4]$	
1	$S_1[3,2]$	$S_4[3,1], S_5[5,3]$ $S_6[5,3]$	
2	$S_4[3,1]$		



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L

R

123456978 = S_8

1	2	3
4	5	6
⑨	7	8

$g(S_7)=3,\ h(S_7)=2,\ f(S_7)=5$

S_7 = 123456789

1	2	3
4	5	6
7	8	⑨

$g(S_8)=3,\ h(S_8)=0,\ f(S_8)=3$

123496758

1	2	3
4	⑨	6
7	5	8

(level 3)

(Already Visited)

Generate every legal successor of S_0

$$newCost \leftarrow g(S_0) + 1 = 0 + 1 = 1.$$

$parent[V] \leftarrow S_0$

$g(V) \leftarrow newCost = 1$

V: S_1	S_2	S_3
$g(S_1) = 1$	$g(S_2) = 1$	$g(S_3) = 1$
$h(S_1) = 2$	$h(S_2) = 4$	$h(S_3) = 4$
$f(S_1) = 3$	$f(S_2) = 5$	$f(S_3) = 5$

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move $(s_1) = R$ move $(s_2) = D$ move $(s_3) = U$

$h(s_1) = d_5 + d_8 = 1$

$h(s_2) = d_4 + d_5 + d_7 + d_8 = 1 + 1 + 1 + 1 = 4$

$h(s_3) = d_4 + d_5 + d_7 + d_8 = 1 + 1 + 1 + 1 = 4$

Dequee Min: $S_1[3,2]$

$$\text{NewCost} = g(S_1)+1 = 1+1=2$$

	L	R	D	U
V:	$S_{visited}$	S_4	S_5	S_6

$\downarrow$

$S_{vis}=S_0 \in$ g-table

$g(S_{vis})=g(S_0)=0 \not> \text{newCost}=2$

$\therefore$ therefore S_{vis} is not updated.

$\parallel$

S_0

S_4	S_5	S_6
$g = 2$	$g = 2$	$g = 2$
$h = 3$	$h = 1$	$h = 3$
$f = 5$	$f = 3$	$f = 5$

$$h(S_4) = d_6 + d_5 + d_8 = 3$$

$$h(S_5) = d_8 = 1$$

$$h(S_6) = d_2 + d_5 + d_8 = 3$$

Dequeue Min:

$S = S_4[3, 1]$

$newCost \leftarrow g(S_4) + 1 = 2 + 1 = 3$

v:	R	L	U
	S_7	S_8	$S_9 = S_{vis}$

g-table :

State (S_i)	$g(S_i)$	$h(S_i)$	$f(S_i)$	Parent (S_i)	Move (S_i)	Sorted Priority Q for new nodes
S_0	0	3	3	Nil	Nil	$S_1[3,2]$, $S_2[5,4]$, $S_3[5,4]$
S_1	1	2	3	S_0	R	$S_4[3,1]$, $S_5[5,3]$, $S_6[5,3]$
S_4	2	1	3	S_1	D	

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